

Introduction

Markerless motion capture (MMC) accuracy is fundamentally limited by **3D body coverage** and **multi-view redundancy**.

- Suboptimal placement induces joint angle errors up to **20°** [1].
- Poor spatial distribution reduces keypoint visibility and triangulation robustness, especially during fast or occluded actions [2].
- Lab camera placement remains **largely manual and intuition-based** — undetected blind spots and unbalanced bilateral coverage.

Aim of this pilot study

Compare an expert-designed manual setup against the configuration produced by our open-source **Lab Camera Optimizer** in the same lab, with an identical 9-camera mixed inventory (7 ZED2i + 2 iPad).

Methods

Composite score per point

- **Vertical body coverage** — penalty v_i^2 , gate $v_i \geq 0.9$
- **Distance attenuation** — $\frac{1}{1+kd_i^2}$ far cameras count less.
- **Angular diversity** — near-duplicate views ($\Delta\theta < 20^\circ$) get partial credit.
- **Saturated coverage** — diminishing returns above target N_t : $base(p) = \frac{1}{\sqrt{N_t}} \cdot \sum_i S_i^{(i)} / \sqrt{i}$
- **Bilateral factor** — geometric balance south/north of the walking axis: $B(p) = \frac{\sqrt{S \cdot N}}{\max(S, N)}$

Per-point score

Per-camera contribution — $s_i = v_i^2 \cdot \frac{1}{1+kd_i^2}$

Per-point score — $f(p) = w_p \cdot base(p) \cdot [(1 - \beta) + \beta B(p)]$

Layout score (maximised) — $F = \sum_p f(p)$

Parameters : $k = 0.001$; $\beta = 0.8$; $N_t = 6 \text{ cam/point}$; S, N : saturated sums south/north of walking axis

Results

+58 %

Global coverage score

576.0 (algo) vs 363.8 (manual)

100 % vs 28 %

Zone meeting ≥ 6 -camera target

redundancy across analysis zone

4 L + 3 P

Mixed orientations

vs 7 L + 0 P (manual)

L = Landscape ; P = Portrait

Room

+ obstacles

+ FOV

~1145 candidates

(walls $\times$ heights

$\times$ orientations)

311 eval points

on sample grid

Farthest-point init

→ Greedy 1-opt

$\times 100$ restarts

Best layout

Score 576.0

Experts' zone

confirmed optimal

Repeated $\times 6$, one per candidate capture-zone position

(a) Manual setup — Score 363.8 • 7 L + 0 P

(b) Optimised setup — Score 576.0 • 4 L + 3 P

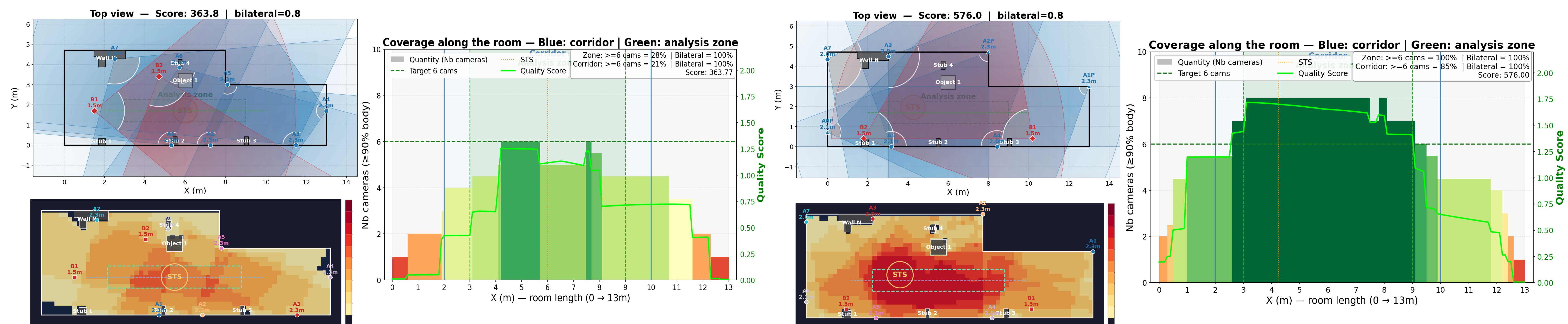


Fig. 1 — (a) Manual vs (b) optimised setup. Each composite: top-left = room plan with camera FOV; bottom-left = XY 3D camera-count heatmap; right = 1D coverage profile along the analysis zone (bars = camera count, green line = quality score, green band = ≥ 6 -camera sub-zone).

Discussion & Conclusion

Where experts win — macro planning: the chosen capture zone is the room's global optimum (best of 6 candidates).

Where experts struggle — micro placement: too many interacting axes (orientation, position, height, pan/tilt, bilateral balance) for intuition alone.

Concrete gain — optimising L/P combinations + line-of-sight raises the ≥ 6 -camera redundancy from **28 %** → **100 %** of the analysis zone, *without extra hardware*.

Next steps — multi-lab replication, joint-angle validation vs marker-based reference, end-to-end co-optimisation of the capture zone.

Open-source tool



github.com/flodelaplace/lab-camera-optimizer
pip install lab-camera-optimizer • MIT